Rate Map

Menu: File > Rate Map

The Rate Map enables variable rate application by defining geographic zones with different target rates. It integrates with AgOpenGPS for GPS-referenced mapping.

Prescriptions Tab

Manages prescription shapefiles for the current field. The active prescription determines which zone rates are used during application.

Control	Action
Field	Shows the field name assigned to the current job
Prescription list	Lists all prescription files available for the current field. Select one to make it active. **Zones.shp** is the default and cannot be deleted.
New	Create a new empty prescription file with a given name
Copy	Copy the selected prescription to a new name. The copy becomes the active prescription.
Import	Open the prescription import dialog — imports a shapefile or an AgGrow XML prescription. See Importing a Prescription below.
Export	Export the selected prescription shapefile to a chosen location
Delete	Delete the selected prescription file. Zones.shp cannot be deleted.

> **Note:** Selecting a prescription from the list immediately makes it the active prescription for the current job. The map reloads to show the selected zones.

Importing a Prescription

The Import button opens a two-step import dialog:

Step 1 — Import Data

Control	Description
Shape file / XML	Choose the source format: a prescription **shapefile** (.shp) or an **AgGrow XML** prescription file
Import	Select the file and read it in
Mapping grid	For shapefiles, match each product (A–E) to the attribute column in the shapefile that holds its rate. For XML files, the layer names found in the file are shown.
Product	The primary product (A–E) used when adjusting zones in Step 2. Selected automatically from the imported data; change it if needed.
Name	File name for the saved prescription

Step 2 — Adjust Zones (optional)

Grid-style prescriptions can contain thousands of tiny zones. Adjust consolidates them into fewer, larger zones that are easier to work with and apply.

Setting	Description
Max Zones	Maximum number of distinct rate zones to create
Min zone size	Zones smaller than this area (Ac/Ha) are merged into their neighbours
Min zone step	Minimum rate difference between zones — rates closer together than this are combined
Adjust	Run the consolidation and report how many zones were created

Press **Save** to save the prescription for the current field. It becomes the active prescription. If the imported data has different product names, RC offers to update the job's product names to match.

Zones Tab

Create and edit rate zones within the active prescription.

Create zones by clicking the New button, then clicking at least **3 points** on the map to define the boundary.

- With **overlapping zones**, the **last zone added** takes priority.
- When variable rate is enabled but **no zone exists**, the **base rate** is used.

Button	Action
New	Create a new rate zone
Edit	Edit zone properties (name, rate per product, color)
Delete	Delete corner point (while creating) or delete zone (while editing)
Cancel	Cancel current edit
Save	Save current edit
Zone List	Open the Zone List — view all zones with their areas, product rates (A–E), and colors; select/deselect zones; delete zones; print the list

Data Tab

Control	Action
Record count	Number of applied rate data points recorded
A / B / C / D / E	Select which product's applied rate to display on the map
Record	Start/stop recording applied rates
Delete	Delete all recorded applied rate data

VR Tab — Variable Rate Look-Ahead

Configures how far ahead of a zone boundary the application rate begins adjusting, so the rate matches the target as soon as the machine enters the new zone. Compensates for applicator response lag.

Setting	Description
Look-Ahead Time (per product)	Seconds before entering a zone boundary that rate adjustment begins. One value per product (up to 5).
Auto Tune	RC monitors look-ahead performance and automatically adjusts the value to improve zone entry accuracy.

Files Tab

Import and manage supporting data files for the current field. Use the file selector dropdowns to choose which file is active for each overlay layer.

Field Boundary

The field boundary defines the operated area and is used to clip automatically generated productivity zones. Select the active boundary from the **KML** dropdown. Import a boundary using the **Import Boundary** button — three file types are accepted:

- **AOG Field.kml** — the field KML from the AgOpenGPS Fields folder; the boundary polygon is extracted automatically.
- **AOG Boundary.txt** — AgOpenGPS boundary file; easting/northing coordinates are converted to lat/lon using the field origin.
- **Any KML (*.kml)** — any KML file containing a polygon boundary.

> **Tip:** Field files are typically found in:
> - **AgOpenGPS:** Documents\AgOpenGPS\Fields\[Field Name]\ — contains `Field.kml` and `Boundary.txt`
> - **TWOL:** Documents\TWOL\Fields\[Field Name]\ — contains `Boundary.txt`

Yield, EC, and Elevation Data

Control	Action
Yield dropdown	Select the active yield data file for the overlay and zone creation
Import Yield	Import a yield data CSV file. If the file contains elevation data, RC will offer to extract it to a separate elevation file.
Delete Yield	Remove the selected yield data file
EC dropdown	Select the active EC (electrical conductivity) data file
Import EC	Import an EC data CSV file (Lat, Lon, EC)
Delete EC	Remove the selected EC data file
Elevation dropdown	Select the active elevation data file
Import Elevation	Import an elevation data CSV file (Lat, Lon, Elevation)
Restore Elevation	Restore the previous elevation file from the automatic backup
Delete Elevation	Remove the current elevation data file

Create Zones

The **Create Zones** button opens the productivity zone creation dialog. Zones are generated automatically from yield history, soil EC, elevation data, or any combination. Clear existing zones before generating if you want to replace them.

Setting	Description
Yield files (checklist)	Check one or more yield data files to include. Multiple years are blended by their assigned weight.
EC	Include the active EC layer in zone generation
Elevation	Include the active elevation layer in zone generation
Weights	Percentage weight for each active layer. All weights must sum to 100 before the Build button is enabled.
Productivity levels	Number of productivity classes to create (e.g. 3 = Low / Med / High). Each class may produce more than one zone polygon if the field has non-contiguous areas of the same productivity level.
Min zone size	Minimum zone area in hectares (or acres). Zone polygons below this size are discarded. Increase to reduce the number of small fragments.
Build	Generate zones using the selected layers and settings. The field boundary (set in the KML dropdown) is used to clip zones to the operated area.

> **Tip:** Set the field boundary KML before generating zones. Without a boundary, zones are clipped to the bounding box of the data files, which may not match the actual field edge.

Display Toggle Buttons (right panel)

These buttons highlight green when active. Click to toggle on or off.

Button	Action
Sat. View	Toggle satellite background image
Target Zones	Toggle rate zone boundaries display
Applied Rates	Toggle applied rate colour overlay
KML Files	Toggle KML file overlay
Yield Data	Toggle yield data colour overlay
EC	Toggle soil EC colour overlay
Elev.	Toggle elevation colour overlay

Bottom Controls

Button	Action
Minimize	Shrink the map to a small window. Drag by title bar to reposition.
Centre	Centre the map on the current GPS position
Help	Open this help page
OK	Close the Rate Map